

Exploratory Data Analysis

In [1]:

```
import pandas as pd
```

In [2]:

```
df = pd.read_csv("ott_movies_clean_unique.csv")
```

In [3]:

```
df.head()
```

Out[3]:

	content_id	title	type	genre	platform	country
0	C100146	Silent Code 146	Series	Drama	Netflix	UK
1	C100452	Crimson Truth 452	Movie	Action	Netflix	Germany
2	C101169	Cyber Night 1169	Series	Thriller	Prime Video	Germany
3	C100271	Silent Code 271	Movie	Sci-Fi	Netflix	India
4	C100591	Silent Code 591	Movie	Drama	Hotstar	Germany

In [4]:

```
df.shape # Number of rows & Columns
```

Out[4]:

(2500, 19)

In [5]:

```
df.columns # Column names
```

Out[5]:

```
Index(['content_id', 'title', 'type', 'genre',  
      'platform', 'country',  
      'language', 'release_year', 'duration_m  
inutes', 'rating', 'votes',  
      'weighted_rating', 'engagement_score',  
      'popularity_score',  
      'trending_score', 'tags', 'combined_fea  
tures', 'description',  
      'poster_url'],  
      dtype='object')
```

In [6]:

```
df.info() # Data types
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2500 entries, 0 to 2499
Data columns (total 19 columns):
#   Column                Non-Null Count  Dtype
---  -
0   content_id            2500 non-null   object
1   title                 2500 non-null   object
2   type                  2500 non-null   object
3   genre                 2500 non-null   object
4   platform              2500 non-null   object
5   country               2500 non-null   object
6   language              2500 non-null   object
7   release_year          2500 non-null   int64
8   duration_minutes      2500 non-null   int64
9   rating                2500 non-null   float64
4
10  votes                 2500 non-null   int64
11  weighted_rating       2500 non-null   float64
4
12  engagement_score      2500 non-null   float64
4
13  popularity_score      2500 non-null   float64
4
14  trending_score        2500 non-null   float64
4
15  tags                  2500 non-null   object
16  combined_features     2500 non-null   object
17  description            2500 non-null   object
18  poster_url            2500 non-null   object
dtypes: float64(5), int64(3), object(11)
memory usage: 371.2+ KB
```

In [7]:

```
# check missing values  
df.isnull().sum().sort_values(ascending=False)
```

Out[7]:

content_id	0
votes	0
description	0
combined_features	0
tags	0
trending_score	0
popularity_score	0
engagement_score	0
weighted_rating	0
rating	0
title	0
duration_minutes	0
release_year	0
language	0
country	0
platform	0
genre	0
type	0
poster_url	0
dtype:	int64

In [8]:

```
# Statistical Summary
df.describe()
```

Out[8]:

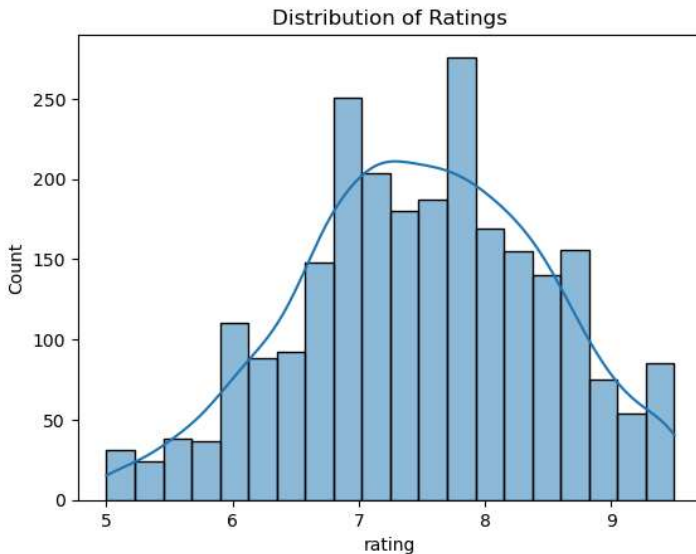
	release_year	duration_minutes	rating	
count	2500.000000	2500.000000	2500.000000	250
mean	2019.578800	634.919600	7.487160	10046
std	2.861864	617.055317	0.980232	5744
min	2015.000000	80.000000	5.000000	52
25%	2017.000000	128.000000	6.800000	5109
50%	2020.000000	302.000000	7.500000	10068
75%	2022.000000	1116.000000	8.200000	15019
max	2024.000000	2000.000000	9.500000	19981

In [9]:

```
# Univariate Analysis
import seaborn as sns
import matplotlib.pyplot as plt
```

In [10]:

```
# Rating Distribution
sns.histplot(df['rating'], bins=20, kde=True)
plt.title("Distribution of Ratings")
plt.show()
```



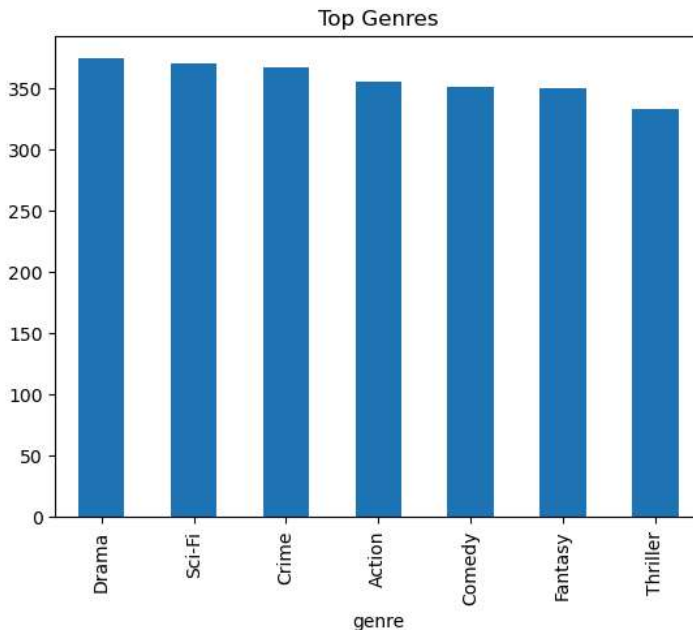
This distribution peaks around 7 to 8. There are some movies in the 5 to 7 range, the highest concentration is slightly above average (around 7-7.5). So most of the movies are above average rather than just average.

There is a strong presence of ratings between 7 and 9. The right side (7.5–9) has many bars with high counts, meaning a significant number of movies are well-rated. However, extremely high ratings (above 9) are rare.

In [11]:

```
# Genre Analysis
```

```
df['genre'].value_counts().head(10).plot(kind='bar')  
plt.title("Top Genres")  
plt.show()
```



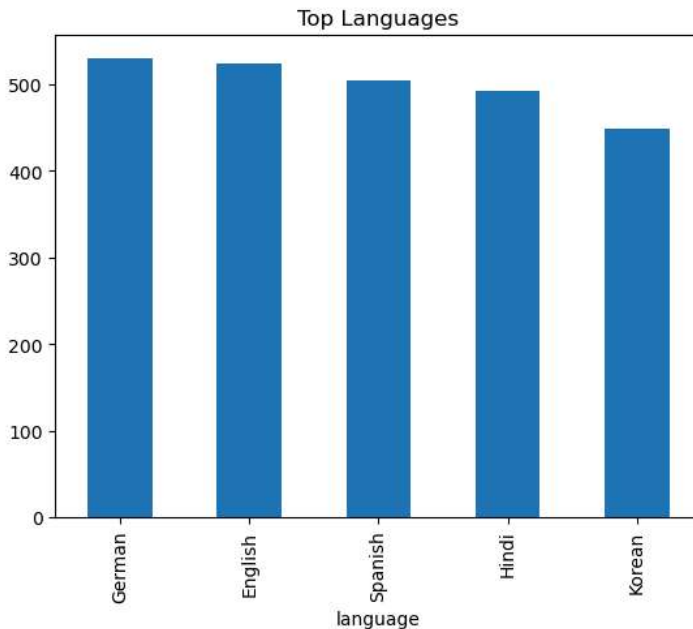
Yes, OTT platforms are heavily dominated by Drama, Sci-Fi, and Crime content. This suggests:

Viewers prefer story-driven and emotionally engaging content (Drama)

High demand for imaginative/tech-based content (Sci-Fi) Strong interest in suspense and investigation (Crime)

In [12]:

```
# Language Distribution
df['language'].value_counts().head(10).plot(kind='bar')
plt.title('Top Languages')
plt.show()
```



According to Language distribution German Language is highest Distribution after that English is very nearly to German Language. Third domination language is Spanish and fourth domination of language is Hindi. Spanish and Hindi are lower distribution than German and English but higher distribution than Korean language.

Bivariate Analysis (relationships)

In [13]:

```
# Rating vs Genre
df.groupby('genre')['rating'].mean().sort_values(ascending=False)
```

Out[13]:

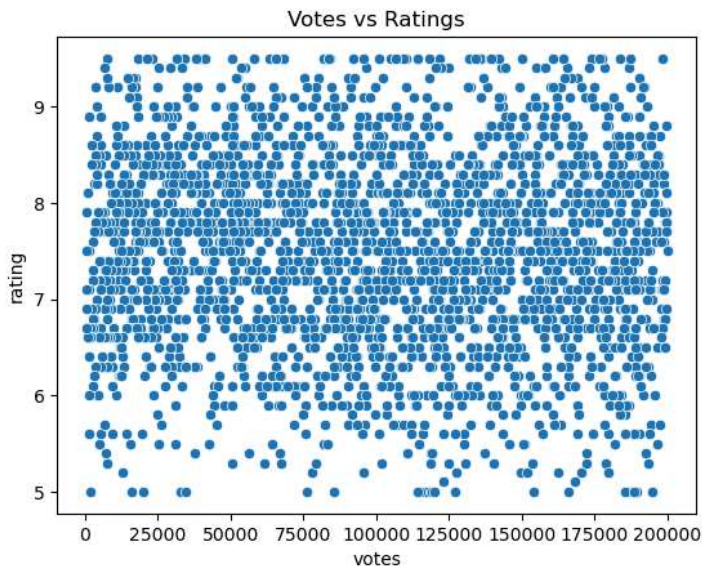
```
genre
Action      7.523944
Sci-Fi      7.515946
Fantasy     7.490286
Drama       7.487701
Comedy      7.477493
Crime       7.476294
Thriller    7.434234
Name: rating, dtype: float64
```

According to Rating vs Genre relationships Action is the highest ratings.
Second highest ratings is Sci-Fi and Lowest ratings is Thriller.

In [14]:

```
# Votes vs Rating
```

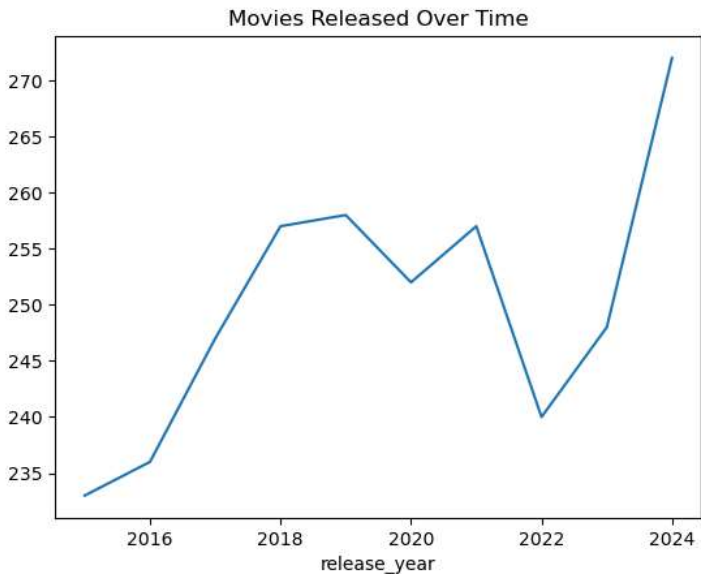
```
sns.scatterplot(x='votes', y='rating', data=df)  
plt.title('Votes vs Ratings')  
plt.show()
```



“Based on scatter plot analysis, there is no strong correlation between votes and ratings. Ratings are distributed across all vote levels, indicating that popularity does not guarantee higher user satisfaction.”

In [15]:

```
# year-wise Trend  
df['release_year'].value_counts().sort_index().plot()  
plt.title("Movies Released Over Time")  
plt.show()
```



Growth of OTT content is enhancing from 2016 to 2024. In 2016 the growth is starting and in 2024 the growth of OTT content is enhancing tremendously.

Feature Engineering

In [16]:

```
df['popularity']=df['rating']*df['votes']
df['description_length']=df['description'].apply(lambda x:len(x))
```

In [18]:

```
df.head()
```

Out[18]:

	content_id	title	type	genre	platform	country
0	C100146	Silent Code 146	Series	Drama	Netflix	UK
1	C100452	Crimson Truth 452	Movie	Action	Netflix	Germany
2	C101169	Cyber Night 1169	Series	Thriller	Prime Video	Germany
3	C100271	Silent Code 271	Movie	Sci-Fi	Netflix	India
4	C100591	Silent Code 591	Movie	Drama	Hotstar	Germany

5 rows × 21 columns



NLP EDA

In [19]:

```
from sklearn.feature_extraction.text import CountVectorizer

cv = CountVectorizer(stop_words='english')
word_matrix = cv.fit_transform(df['description'].astype(str))

word_counts = word_matrix.toarray().sum(axis=0)
words = cv.get_feature_names_out()

word_freq = pd.DataFrame({'word': words, 'count': word_counts})
word_freq.sort_values(by='count', ascending=False).head(20)
```

Out[19]:

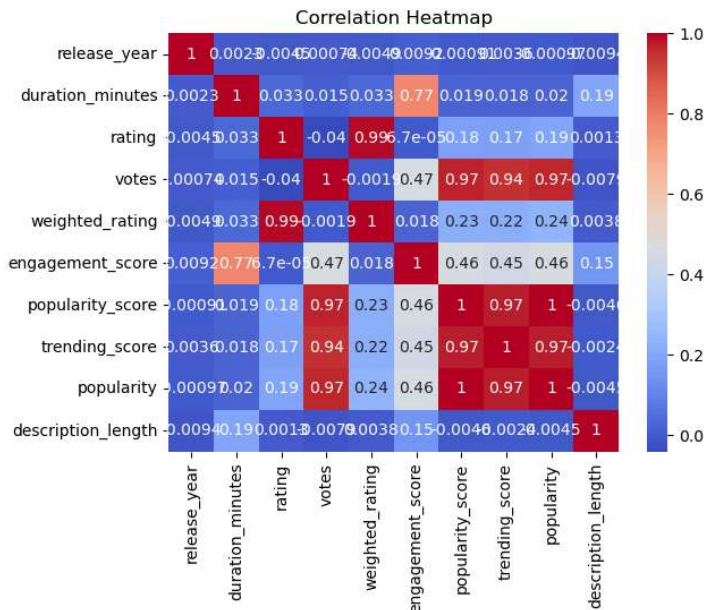
	word	count
2502	engaging	2500
2505	featuring	2500
2510	intense	2500
2526	storytelling	2500
2492	characters	2500
2522	series	1254
2514	movie	1246
2531	video	852
2518	prime	852
2509	hotstar	824
2516	netflix	824
2499	drama	374
2521	sci	370
2506	fi	370
2495	crime	367
2490	action	355
2494	comedy	351
2504	fantasy	350
2528	thriller	333
2500	echoes	233

Engaging, Featuring, Intense, Storytelling, Characters Content dominates

Correlation Heatmap

In [20]:

```
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```



Weighted_rating is the strongest predictor of movie ratings. Audience-related metrics such as popularity, engagement, and trending score show a mild positive influence. Features like release year, duration, and description length do not significantly affect ratings.

In [21]:

```
df.to_csv('new_cleaned_data.csv', index=False)
```

In [22]:

```
df.head()
```

Out[22]:

	content_id	title	type	genre	platform	country
0	C100146	Silent Code 146	Series	Drama	Netflix	UK
1	C100452	Crimson Truth 452	Movie	Action	Netflix	Germany
2	C101169	Cyber Night 1169	Series	Thriller	Prime Video	Germany
3	C100271	Silent Code 271	Movie	Sci-Fi	Netflix	India
4	C100591	Silent Code 591	Movie	Drama	Hotstar	Germany

5 rows × 7 columns



In []:

In []: